

Please write your name here

MASTER 2 – TRACK: EBDS 2024

Multivariate and Nonlinear Time Series

Final exam

Thursday 22 January 2026

- Students are permitted to bring into the examination room: any document, book they want
- Students are NOT permitted to use during the examination mobile phones and to communicate with each other.

Instructions

- Write your **first name and surname** in the space provided above on this page.
- All written responses must be in English.

At the end of the examination

- Place your file with the answers on AMETICE

Time Series

Final examination

Exercise 1. – Modelling French unemployment rate using regime-switching models

Download data from the excel file "unemploy-france". This file contains unemployment rate data for France, at monthly frequency, since 1960.

Question 1.- Estimate the **best ARMA model** on the time series. Plot the histogram and the density of the residuals. Is the estimated model satisfactory? Apply several **residual diagnostic tests** and comment.

Question 2. Linearity tests

Apply **linearity tests against two-regime SETAR and LSTAR models**, with the transition variable being lagged unemployment rates (choose several lags up to the fourth lag. What is your conclusion?

Question 3: Estimation of LSTAR models

a) Estimate the **best LSTAR model**. Consider different lags for the transition variables (lagged unemployment rate) and use residual diagnostic test to select the most appropriate model. Interpret your results.

b) Plot the fitted time series, **the transition function** and any other graph you find useful

Exercise 2. – Modelling extreme distributions

Data and variables

Download the Excel file `extremes.xlsx`. It contains **two daily time series** (one observation per day) covering **1990–2019**:

Sheet `daily` includes:

- `date` the calendar date (daily frequency)
- `tmax_c` daily **maximum temperature** in **degrees Celsius**
- `precip_mm` daily **precipitation** in **millimeters**
- `year, month` (provided for convenience)

Question 1: Load, inspect, and visualize the data

1. Import the Excel file into R and convert `date` to a proper `Date` object.
2. Plot `tmax_c` and `precip_mm` versus time.
3. Write a short description of what you see: seasonality, persistence (serial correlation), occasional spikes (potential extremes).

Question 2 : Block maxima method

We treat each year as one block.

1. Compute the **annual maximum** of `tmax_c` and `precip_mm` for each year.
2. Plot annual maxima versus year.
3. Fit a **GEV** model to the annual maxima using maximum likelihood.
4. Produce standard diagnostic plots (QQ plot / probability plot / return level plot).
5. Write a short paragraph interpreting the estimated tail type based on the sign of the shape parameter

Question 3 : POT method

1. Choose a high threshold u for `tmax_c` and `precip_mm` (start around the 0.95 quantile).
2. Fit a **GPD** to exceedances over u .
3. Generate diagnostic plots.
4. Estimate return levels for 10, 50, and 100 years.
5. Briefly discuss how sensitive results are to the chosen threshold (try at least two thresholds, e.g. 0.95 and 0.99).

Question 3 : Compare Block Maxima vs POT

Compare the two approaches in terms of:

- How much data they use (annual maxima uses 1 value per year; POT uses many exceedances)
- Sensitivity to block size vs threshold choice
- Similarities/differences in estimated return levels

Exercise 3. – Copula

Download the Excel file `spsent.xlsx` (sheet `DIV`). It contains **two series observed at monthly frequency from 2000-01 to 2024-12**:

- `sp500_index` a simulated stock market index level
- `sentiment_index` a simulated investor sentiment index level

In the applied part, you will study the dependence between **monthly stock returns** and **the sentiment cycle** (you will decide whether to work with sentiment levels or sentiment changes, and justify your choice briefly).

Question 1. What is an Archimedean copula?

- a) Explain what an **Archimedean copula** is, starting from the idea of a **generator** φ . Your answer should include:

- the general form $C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$ (2D case)
- the key conditions on φ (monotone, etc.) in plain language
- one practical advantage of Archimedean copulas (compared to, e.g., elliptical copulas) and one limitation

Question 2. Compare three Archimedean copulas by simulation ($n = 1000$)

Simulate **1000 observations** from each of the following bivariate copulas:

- **Clayton** (lower-tail dependence)
- **Gumbel** (upper-tail dependence)
- **Frank** (no tail dependence, more “middle” dependence)

For each one:

- choose a parameter that gives a clear dependence (not too weak, not too strong)
- simulate (U_1, U_2) on $[0, 1]^2$
- draw the scatterplot of U_1 vs U_2
- comment on how the scatterplot reflects **asymmetry** and **tail dependence**

What to submit: 3 scatterplots and a short comparison paragraph (Clayton vs Gumbel vs Frank).

Question 3 — Tail dependence coefficients (compute and interpret)

For **Clayton** and **Gumbel**, recall (or find) the theoretical expressions of:

- the **lower tail dependence coefficient** λ_L
- the **upper tail dependence coefficient** λ_U

Then:

- compute λ_L and λ_U for your chosen parameters
- explain in plain English what these numbers mean in terms of “joint extremes”

What to submit: formulas, numerical values, and 5-8 lines of interpretation.

Question 4. Archimedean generator

For **one** of the three copulas you simulated (Clayton, Gumbel, or Frank):

- write the generator $\varphi(t)$
- write the resulting copula formula $C(u, v)$
- briefly explain how the parameter affects dependence strength

Question 5. Fréchet–Hoeffding bounds (numerical check)

Recall the Fréchet–Hoeffding bounds:

- $W(u, v) = \max(u + v - 1, 0)$
- $M(u, v) = \min(u, v)$

Take **three observed pairs** (u_i, v_i) from your **Clayton** simulated sample (pick any 3 rows). For each pair:

- compute $W(u_i, v_i)$, $C(u_i, v_i)$, $M(u_i, v_i)$
- verify numerically that $W \leq C \leq M$

What to submit: a small table with the three points and the three values W, C, M , plus one sentence confirming the inequality.

Question 6. Construct the two series to be linked

Using the Excel file:

a) Compute the monthly **percentage return** (or log-return) of the index:

- percentage return: $r_t = 100 \times (P_t/P_{t-1} - 1)$
- log return: $r_t = \log(P_t) - \log(P_{t-1})$

b) Decide how to treat sentiment:

- either use sentiment_index (levels)
- or use changes $\Delta s_t = s_t - s_{t-1}$

Justify your choice in 3-5 lines (stationarity, cycles, etc.).

Question 7. Study the marginal distributions

For the two final series (returns and your sentiment measure):

- plot histograms and QQ-plots
- propose a reasonable marginal model for each (examples: Normal, Student-t, skewed-t, exponential, gamma, or empirical distribution)

Question 8. Fit and compare Archimedean copulas

Convert the data to pseudo-observations (U, V) (ranks / empirical CDF). Then fit and compare at least the following copula families:

- Clayton
- Gumbel
- Frank

You can use `VineCopula::BiCopSelect()` but **restrict the candidate set** to Archimedean families only (Clayton/Gumbel/Frank). Report:

- the selected family
- the estimated parameter(s)
- Kendall's tau (or another dependence measure) implied by the fit

Question 9. Simulate from your fitted copula and compare to data

Using the fitted Archimedean copula:

- simulate 275 observations (U^*, V^*)
- plot the scatterplot
- compare visually to the empirical pseudo-observations from the real data

Question 10. Which copula does this look like?

Based on:

- your scatterplots
- the estimated tail dependence behavior (if any)
- the selected family

Answer:

- which Archimedean copula does your dependence structure resemble most
- write its formula $C(u, v)$
- explain in 4–6 lines why it makes sense here (or why it might be questionable)

Exercise 4. – Markov-switching modelling of market sentiment

We continue using one the time series in DIV database. Specifically, one would like to adjust a two-regime Markov-switching model to the series of the log returns of S&P 500

1.-Estimate the best ARMA model with constant intercept, considering a maximum lag of 4 for both the AR and MA component. Apply the usual diagnostic tests on the residuals. Comment.

2.- Estimate a Markov-switching regime with 2 regimes.

3.- Do you find the estimated model satisfactory? If no, What could be done to improve the specification?

4 -Show a graph with the initial series of sentiments and regime 1 or 2. Comment.